9.1 In der Übungsstunde behandelt

9.2

m

$h = 3R$

θ

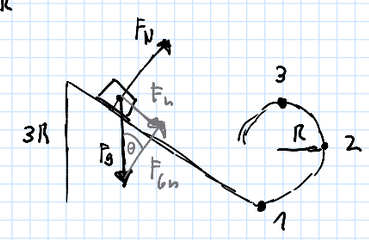
$v_0 = 0 \frac{m}{s}$

$v_1 = \sqrt{6gR}$

$v_2 = \sqrt{4 \cdot g \cdot R}$

$v_3 = \sqrt{2 \cdot g \cdot R}$

(a)



(b) $F_n = \sin(\theta) \cdot F_G$

$F_{Gn} = \cos(\theta) \cdot F_G$

(c) $a = \sin(\theta) \cdot g$

$\frac{1}{2} a t^2 = \Delta s$

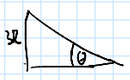
(d) Wir kennen die Folgenden Gleichungen

$\frac{1}{2} \sin(\theta) \cdot g \cdot t^2 = \Delta s$

$\sin(\theta) \cdot g = a$

$v = a \cdot t$

$\sin(\theta) = \frac{3R}{\Delta s}$



$\Delta s = \frac{3R}{\sin(\theta)}$

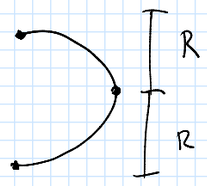
$t^2 = \frac{2 \Delta s}{\sin(\theta) \cdot g}$

$t = \sqrt{\frac{2 \cdot 3R}{\sin^2(\theta) \cdot}}$

$v = a \cdot t = \sin(\theta) \cdot g \cdot t = \sqrt{\frac{\sin^2 \cdot g^2 \cdot 6R}{\sin^2 \cdot g}}$

$v = \sqrt{6gR}$

(e)

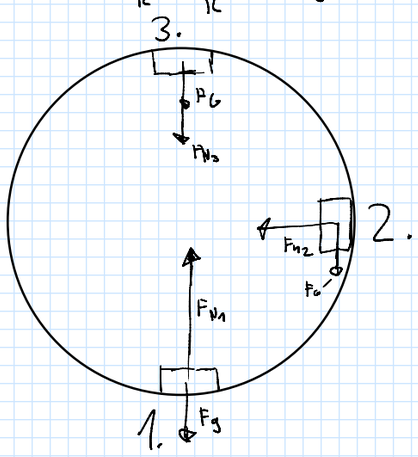


1. $\frac{v_1^2}{R} = \frac{6gR}{R} = a_1 = 6g$

2. $\frac{v_2^2}{R} = \frac{4gR}{R} = a_2 = 4g$

3. $\frac{v_3^2}{R} = \frac{2gR}{R} = 2g$

(f)



(h) $\frac{v_3^2}{R} = g$

$v_3 = \sqrt{\frac{g}{R}}$